Erik Kalz

Curriculum Vitae

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ErikKalz

Education

2022 -2025 **Doctorate in theoretical physics**, *University of Potsdam*, Supervisors: Prof. Ralf Metzler and Prof. Carsten Beta

2020 -2022 **Master of physics**, *Technical University of Dresden*, Supervisors: Prof. Jens-Uwe Sommer and Dr. Abhinav Sharma. Minor: philosophy

2016 -2020 **Bachelor of physics**, *Technical University of Dresden & Norwegian University of Science and Technology, Trondheim*, Supervisors: Prof. Jens-Uwe Sommer and Dr. Abhinav Sharma. Minor: philosophy

2016 **Abitur**, *Max-Steenbeck Gymnasium Cottbus*, Abitur with best mark 1.0

Awards

Michelson Price: I was awarded the "Michelson Price" of the Faculty of Science at the University of Potsdam for best doctoral thesis of the year 2025.

IOP Impact Award: The publication "Field theory of active chiral hard disks: a first-principles approach to steric interactions" in *Journal of Physics A* has been recognised for the "impact [it has] achieved in such a short period of time", Ref. [**B5**].

Emergent Talent: Awarded to my talk "Subtle interactions in odd-diffusive systems" at the conference "Venice meeting on fluctuations in small complex systems VII" (2024) in Venice, Italy.

Best Communication: Awarded to my talk "Interactions enhance self-diffusion in odd-diffusive systems" at the conference "New Trends in Nonequilibrium Statistical Mechanics" (2023) in Erice, Sicily.

Springer BestMasters: Springer Nature awards publication to the best master's theses which have been completed at renowned Universities in Germany, Austria, and Switzerland (2022), Ref. [**C1**].

PRL Editors' Suggestion: The publication "Collisions enhance self-diffusion in odd-diffusive systems" in *Physical Review Letters* was selected as an "Editors' Suggestion" (2022), Ref. [**B7**].

Academic Metrics & Services

Publications: 1 preprint under review, 8 peer-reviewed journal articles (4 first-authored, 3 as corresponding author), 1 book

Citations: 210+ citations, h-index: 7 (Google Scholar)

Conferences: 20+ invited talks, 10+ contributing talks, and 5+ contributing posters at international conferences

Organization:
- Guest-editing a *New Journal of Physics* 'Focus On' issue on " Broken symmetries and odd transport in statistical physics" together with R. Metzler and A. Sharma
- DPG-SKM 2025 focus session "Broken symmetries in statistical physics: Dynamics of odd systems" together with R. Metzler and A. Sharma
- 4 semester Metzler group seminar (70+ international speakers)

Supervision: 6 Bachelor students (5 finished, 1 ongoing), 3 Master students (1 finished, 2 ongoing)

Teaching
- undergraduate teaching assistant: 1 semester mathematics for physicists, 4 semesters mathematics for engineers, 4 semesters logic for philosophy,
- graduate teaching assistant: 1 semester theoretical mechanics for physicists
- lecture: "Nonequilibrium Statistical Physics" jointly shared with R. Metzler

Education

Dissertation *Tracer transport in interacting odd-diffusive systems.* Supervised by Prof. Ralf Metzler and Prof. Carsten Beta. The cumulative dissertation explores the microscopic and mesoscopic dynamics of systems with broken time-reversal and parity symmetries, with a particular focus on odd-diffusive (also chiral) systems. Through a combination of first-principles theory, kinetic approaches, and statistical mechanics in the dilute limit, as well as field-theoretic approaches for crowded systems, I show how such symmetries, or their deliberate breaking, manifest themselves in unconventional transport phenomena of tracer particles. Contributing publications are Refs. [**B6**, **B5**, **B4**, **B3**]. The thesis was graded with **summa cum laude** ("highest distinction") and awarded the "**Michelson Price**" for the best doctoral thesis in the year 2025 at the Faculty of Science at the University of Potsdam.

09/2022 -12/2025 **Doctorate in theoretical physics**, *University of Potsdam*, Germany

Master Thesis *Diffusion under the effect of Lorentz force.* Supervised by Prof. Jens-Uwe Sommer and Dr. Abhinav Sharma. Brownian particles under the effect of Lorentz force show an unexpected diffusive behaviour: Collisions can enhance the self-diffusion instead of reducing it, as ordinarily. The thesis was graded with **best mark 1.0** and published in the Springer-Nature book-series **BestMasters**, Ref. [**C1**]. The research was published as a *Phys. Rev. Lett.*, Ref. [**B7**], and selected as an **Editors' Suggestion**.

04/2020 -05/2022 **Master of physics**, *Technical University of Dresden*, Germany, Minor: philosophy

07/2019 -12/2019 **semester abroad**, *Norwegian University of Science and Technology*, Trondheim, Norway, enrolled as master student

Bachelor Thesis *Entropy production in a non-equilibrium system of hard rods in confinement.* Supervised by Prof. Jens-Uwe Sommer and Dr. Abhinav Sharma. Density Functional Theory was used in equilibrium and in dynamics to study the entropy production of interacting particles. The thesis was graded with **best mark 1.0**

10/2016 -01/2020 **Bachelor of physics**, *Technical University of Dresden*, Germany, Minor: philosophy

07/2016 **Abitur with best mark 1.0**, *diploma from german secondary school, qualifying for university admission or matriculation*

2010 -2016 **secondary school**, *Max-Steenbeck-Gymnasium Cottbus*, Germany, *secondary school with extended education in mathematics, science, computer science and technology*

Experience

teaching

09/2022 -03/2023 **graduate teaching assistant**, *University of Potsdam*, Germany
- tutor at the Institute for Physics and Astronomy, tutoring the lecture "classical mechanics" during ST 2023

10/2018 -03/2022 **undergraduate teaching assistant**, *Technical University of Dresden*, Germany

- tutor at the Institute of Analysis, tutoring the lecture "advanced analysis for physicists" during WT 2021/2022
- tutor at the Institute of Numerical Mathematics, tutoring the lecture "mathematics for engineering" in WT 2018/2019, ST 2019, ST 2020 and WT 2020/2021
- tutor at the Institute of Philosophy, tutoring the lecture "foundations of logic" in WT 2018/2019, WT 2019/2020, WT 2020/2021 and WT 2021/2022

working

since 09/2022 **graduate research assistant**, *Institute for Physics & Astronomy, University of Potsdam*, Germany

06/2022 -03/2023 **assistance of tax consultance**, *Agency of Ramona Kalz*, Finsterwalde, Germany
Tax-consulting assistance in the application of the renewed German property tax law.

09/2021 -02/2022 **script-writing**, *Institute of Analysis, Technical University Dresden*, Germany
Lecture notes for the course "advanced analysis for physicists" by Priv.-Doz. Dr. A. Kalauch

07/2020 -03/2022 **undergraduate research assistant**, *Leibniz-Institut for Polymer Research Dresden*, Dresden, Germany
Analytical and numerical calculations in the statistical physics of active soft matter in the group of Dr. Abhinav Sharma

summer 2014 **life-guard**, *at open air bath*, Finsterwalde, Germany

miscellaneous

Drivers license: class B, German classification

Languages: **German**: mothertongue, **English**: fluent, C1

Interests: reading, board-games, swimming (life-guard license: silver), cooking, outdoor, biking, diving (CMAS**)

Organisation

11/2025 **New Journal of Physics 'Focus On'**, *Guest-editing a New Journal of Physics 'Focus On' issue*, "Broken symmetries in statistical physics: Dynamics of odd systems" together with R. Metzler and A. Sharma

03/2025 **DPG-SKM focus session**, *DPG-SKM 2025 ("German Physics Society - Section Condensed Matter") conference in Regensburg, Germany*, "Broken symmetries in statistical physics: Dynamics of odd systems" together with R. Metzler and A. Sharma, invited speakers: Prof. Hartmut Löwen, Prof. Friederike Schmid, and Dr. Cory Hargus

since 03/2024 **Seminar "Theoretical Phyics"**, *Metzler group seminar at the University of Potsdam*, 50+ international speakers

Supervision of students

since 12/2025 **Master project** of Juri Schubert at the University of Potsdam.

since 10/2025 **Master project** of Johannes Birkenmeier at the University of Potsdam.

07/2024 –08/2025 **Master project** of Magdalena Unterreiner at the University of Heidelberg with the title *"From Detection to Dynamics: Machine Learning-based Tracking and Motion Analysis of Vegetative Dictyostelium Discoideum"*, jointly supervised with Dr. Fereydoon Taheri.

03/2025 –02/2026 **Bachelor project** of Thea Enzmann at the University of Potsdam with the title *"Interaction effects on transport coefficients in odd-diffusive systems"*.

04/2024 –05/2025 **Bachelor project** of Florian Moran at the University of Potsdam with the title *"First passage time of an underdamped Brownian particle"*, jointly supervised with Prof. Ralf Metzler.

04/2024 –04/2025 **Bachelor project** of Lena Kuse at the University of Heidelberg with the title *"Deriving a time evolution equation for the one-body probability density function of aligning active chiral particles"*, jointly supervised with Dr. Fereydoon Taheri.

04/2024 –02/2025 **Bachelor project** of Lara Standke at the University of Potsdam with the title *"The odd-diffusive Rouse model"*.

10/2023 –01/24 **Bachelor project** of Amelie Langer at the University of Augsburg with the title: *"Dynamics of an Interacting Odd-Diffusive System"*, jointly supervised with Prof. Abhinav Sharma. Based on the research Ref. [**B4**] was published.

04/2023 –08/2024 **Bachelor project** of Johannes Birkenmeier at the University of Potsdam with the title *"Effects of external drift on odd-diffusive systems"*. Based on the research Ref. [**B2**] is under review.

Conferences, Talks and Posters

06/2026 **Invited talk** at the "LOMA-Laboratoire Ondes et Matiére d'Aquitaine" seminar at the University of Bordeaux, France (invited by Prof. David Dean): *"Unusual transport in odd-diffusive systems"*

06/2026 **Invited talk** at the "Institute Curie" in Paris, France (invited by Prof. Jean-François Joanny): *"Unusual transport in odd-diffusive systems"*

06/2026 **Invited talk** at the "theory club" seminar in the "Laboratoire Jean Perrin" at the Sorbonne University, Paris, France (invited by Ananyo Maitra): *"Unusual transport in odd-diffusive systems"*

06/2026 **Invited talk** at the "StatPhys@PHENIX" seminar in the group of Sophie Marbach at the Sorbonne University, Paris, France: *"Unusual transport in odd-diffusive systems"*

05/2026 **Invited talk** at the seminar in the group of Prof. Chase Broederesz at the Vrije University Amsterdam, Netherlands: *"Unusual transport in odd-diffusive systems"*

05/2026 **Invited talk** at the seminar in the group of Dr. Cyriaque Genet and Prof. Thomas Ebbesen at the University of Strasbourg, France: *"Unusual transport in odd-diffusive systems"*

04/2026 **Invited talk** at the CoE "Plasma-Spin-Energy" at the Max Planck Institute of Microstructure Physics in Halle, Germany: *"Odd skyrmions: Enhanced dynamics due to interactions"*

02/2026 **Contributing talk** at the "BioSoft Retreat" 2026 at Tel Aviv University, Israel: *"Unusual transport in odd-diffusive systems"*

11/2025 **Invited talk** at the "Physical Chemistry Seminar" at the Tel Aviv University, Israel (invited by Prof. Yael Roichman): *"Unusual transport in odd-diffusive systems"*

09/2025 **Invited talk** at the "Institute of Spintronics and Quantum Information Seminar" at the Faculty of Physics and Astronomy, Adam Mickiewicz University Poznań, Poland (invited by Prof. Antoni Wójcik & Janek Wójcik): *"Unusual transport in odd-diffusive systems"*

06/2025 **Invited talk** at the "Applied Mathematics Seminar" in the Mathematical Sciences Department, University of Loughborough, UK (invited by Prof. Andrew Archer): *"Unusual transport in odd-diffusive systems"*

06/2025 **Invited talk** at the "Industrial and Applied Mathematics Seminar" at the Mathematical Institute, University of Oxford, UUK (invited by Prof. Maria Bruna and Prof. Jonathan Chapman): *"Unusual transport in odd-diffusive systems"*

06/2025 **Invited talk** at the seminar in the group of Prof. Michael Cates at the Department of Applied Mathematics and Theoretical Physics, University of Cambridge, UK: *"From first principles to the Active Model B+"*

05/2025 **Contributing talk** at the "Leuven school: Basics of nonequilibrium statistical mechanics" 2025 at KU Leuven, Belgium: *"Unusual dynamics in chiral fluids"*

04/2025 **Contributing poster** at the workshop "New Frontiers in Classical Density Functional Theory" at the University of Fribourg, Switzerland: *"Transport in Interacting Chiral Systems"*

03/2025 **Contributing talk** at the DPG-SKM conference 2025 in Regensburg, Germany: *"Odd mobility in interacting particle systems"*

03/2025 **Contributing talk** at the DPG-SKM conference 2025 in Regensburg, Germany: *"Self-diffusion anomalies of an odd tracer in soft-core media"*

10/2024 **Contributing talk** at the conference "Third Infinity 2024" in Göttingen, Germany: *"The subtle nature of interactions in odd-diffusive systems"*

09/2024 **Contributing talk & poster** at the "Venice meeting on fluctuations in small complex systems VII" in Venice, Italy: **@emerging talent speaker**: *"Subtle interactions in odd-diffusive systems"*, poster in cooperation with Dr. Fereydoon Taheri from the University of Heidelberg: *"The Inherent Chirality in Dictyostelium Discoideum Motility: Implications for Cell-Cell Interaction"*

05/2024 **Invited talk** at the SIAM chapter seminar at the University of Potsdam: *"A Geometric Approach to Interacting Particles"*

05/2024 **Invited talk** at the seminar "Statistical Physics and Nonlinear Dynamics" at the Humboldt University Berlin, Germany (invited by Prof. Benjamin Lindner): *"The Subtle Nature of Interactions in Odd-Diffusive Systems"*

04/2024 **Invited talk** at the seminar in the group of Prof. Holger Stark at the Technical University of Berlin, Germany: *"The Subtle Nature of Interactions in Odd-Diffusive Systems"*

03/2024 **Contributing talk** at the DPG-SKM conference 2024 in Berlin, Germany: *"Oscillating autocorrelation functions and their physical implications in equilibrium odd-diffusive systems"*

03/2024 **Contributing talk** at the DPG-SKM conference 2024 in Berlin, Germany: *"From Active Chiral Particles to the Active Model B +"*

02/2024 **Participation** at the PhD-school "The Simulated Parasite" (DFG-SPP 2332) in Jülich, Germany

01/2024 **Contributing talk & poster** at the meeting "Physics of Parasitism 2024" (DFG-SPP 2332) in Würzburg, Germany: *"Odd Diffusion of Entamoeba (?)"*

10/2023 **Contributing talk** at the conference "New Trends in Noneqquilibrium Statistical Mechanics (NES2023)" in Erice, Sicily: *"Collisions enhance Self-Diffusion in Odd Systems"* **@ Selceted for "Best Communication" Award**

09/2023 **Contributing talk** at the workshop "The 4th International Symposium on 'Operational and Stochastic Methods in Fractional Dynamics <MFD23>' " of the Polish Academy of Sciences, Kraków: *"The Physics of Odd Systems"*

07/2023 **Invited talk** at the seminar of DFG-TRR 146 "Multiscale Simulation Methods for Soft Matter Systems" at the University of Mainz, Germany: *"The Physics of Odd Systems"*

05/2023 **Invited talk** at the seminar "Philosophy of Physics" of Priv. Doz. Uwe Scheffler at the Technical University of Dresden, Germany: *"(Ir-) Reversibility in Classical Mechanics"*

04/2023 **Invited talk** at the seminar in the group of Prof. Christine Sellhuber-Unkel at the University of Heidelberg, Germany: *"The Physics of Odd Systems"*

04/2023 **Contributing talk** at the DPG-SKM conference 2023 in Dresden: *"Odd Diffusion of Interacting Particles"*

03/2023 **Contributing poster** at the CECAM-workshop "Emerging colloidal dynamics away from equilibrium. Chiral active systems." in Lausanne, Switzerland: *"Interactions-Enhanced Self-Diffusion in Odd Systems"*

12/2022 **Contributing poster** at the conference "(Post)Modern-Thermodynamics" in Luxembourg: *"Odd Diffusion of Interacting Particles"*

05/2022 **Invited talk** at the group of Prof. Joseph Brader at the University of Fribourg, Switzerland: *"Collisions increase self-diffusion in odd-diffusive systems"*

04/2022 **Invited talk** at the seminar "Theoretical Physics" in the group of Prof. Ralf Metzler at the University of Potsdam, Germany: *"Collisions increase self-diffusion in odd-diffusive systems"*

04/2022 **Invited talk** at the seminar "Soft Matter" in the group of Prof. Hartmut Löwen at the Heinrich-Heine University of Düsseldorf, Germany: *"Collisions increase self-diffusion in odd-diffusive systems"*

Publications

ORCID: 0000-0003-3294-7365, * These authors contributed equally

A Preprints

A1 J. Wójcik and **E. Kalz**. The chiral random walk: A quantum-inspired framework for odd diffusion. *arXiv preprint*, arXiv:2602.09920, 2026

B Journal Articles

B1 F. Faedi, **E. Kalz**, R. Metzler, and A. Sharma. A mobility based approach to transport in chiral fluids. *New Journal of Physics*, 28:063501, 2026

B2 **E. Kalz***, S. Ravichandir*, J. Birkenmeier, R. Metzler, and A. Sharma. Reversal of tracer advection and Hall drift in an interacting chiral fluid. *Physical Review E*, 113:L042104, 2025 **@open access**

B3 P. L. Muzzeddu*, **E. Kalz***, A. Gambassi, A. Sharma, and R. Metzler. Self-diffusion anomalies of an odd tracer in soft-core media. *New Journal of Physics*, 27(3):033025, 2025 **@open access**

B4 A. Langer, A. Sharma, R. Metzler, and **E. Kalz**. Dance of odd-diffusive particles: A Fourier approach. *Physical Review Research*, 6(4):043036, 2024 **@open access**

B5 **E. Kalz**, A. Sharma, and R. Metzler. Field theory of active chiral hard disks: A first-principles approach to steric interactions. *Journal of Physics A: Mathematical and Theoretical*, 57(26):265002, 2024 **@open access** **@IOP impact award 2024**

B6 **E. Kalz**, H. D. Vuijk, J.-U. Sommer, R. Metzler, and A. Sharma. Oscillatory force autocorrelations in equilibrium odd-diffusive systems. *Physical Review Letters*, 132(5):057102, 2024

B7 **E. Kalz***, H. D. Vuijk*, I. Abdoli, J.-U. Sommer, H. Löwen, and A. Sharma. Collisions enhance self-diffusion in odd-diffusive systems. *Physical Review Letters*, 129(9):090601, 2022 **@selected as PRL Editors' suggestion**

B8 I. Abdoli, **E. Kalz**, H. D. Vuijk, R. Wittmann, J.-U. Sommer, J. M. Brader, and A. Sharma. Correlations in multithermostat Brownian systems with Lorentz force. *New Journal of Physics*, 22(9):093057, 2020 **@open access**

C Books

C1 **E. Kalz**. *Diffusion under the Effect of Lorentz Force*. Springer Spektrum Wiesbaden, 2022